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ORIGINAL ARTICLE

Pictures instead of survey questions: An experimental investigation of the feasibility of using pictures in a housing survey

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Abstract

Anyone with a smartphone can take pictures anytime anywhere. This opens the opportunity for researchers to collect photos to augment traditional Web survey data. We conducted an experimental survey asking 2700 members of the Dutch LISS panel about their dwelling conditions. Depending on the condition, they were asked to either take several photos of their house or answer a set of survey questions about the same topics. This paper documents the feasibility of collecting pictures instead of answers in a web survey and studies the consequences of using pictures/questions or a choice for components of Total Survey Error. We find that respondents in our study were much more willing to answer survey questions than to take pictures, but this difference depended on the difficulty of the topic. We then compare the quality of the data from the pictures of the heating systems with the survey responses. Here we find that our pictures contain more useful information than the survey responses. Overall, we conclude that asking respondents to take a picture within a survey can be a feasible way to collect high-quality data.

KEYWORDS

photos, pictures, smart survey, smartphone, total survey error

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1 | INTRODUCTION

Smartphones have the potential to transform the nature of survey research (Lessof & Sturgis, 2018). While traditional survey questionnaires consist of a set of open-ended and closed-ended questions, mobile technology enables the application of novel data collection tools that go beyond traditional survey measurements (Link et al., 2014). In general, innovative forms of data collection might improve the accuracy of measurement while decreasing participation burden (Keusch et al., 2019; Mulder & de Bruijne, 2019; Revilla et al., 2019). Moreover, collecting data via smartphones has the potential to decrease the problem of information recall or socially desirable answering (Keusch et al., 2019).

Since smartphone devices contain integrated sensors (Link et al., 2014), researchers now have the opportunity to ask respondents to activate them and perform different tasks in addition to the survey (Wenz et al., 2019). Smartphone devices can be utilised to collect a variety of data, such as participant's geolocations, physical movement or internet usage (Keusch et al., 2019; Link et al., 2014).

The fact that almost all smartphones contain high-quality cameras, presents an opportunity to collect photos during a web survey. Potentially, some survey questions can be substituted by the task of taking a photo (Bosch et al., 2019; Link et al., 2014). This is specifically useful for a survey aiming to investigate objective aspects of a participant's situation which they could find hard to report. For instance, in dietary studies measuring persons' food intake, pictures could augment traditional self-reports on daily food consumption or even replace self-reports (Gemming et al., 2015).

Statistics Netherlands regularly conducts sample surveys about the housing situation of the Dutch population (Netherlands Housing Survey [WoON]). One of the aims is to produce statistics about the characteristics of dwellings in the Netherlands. Some key statistics of interest, such as the condition of the house, the state of the garden, or the energy-efficiency of the house are very hard to ask with survey questions. Pictures can here either augment survey data or substitute them. For example, instead of asking a relatively complex set of questions about the nature of the outdoor space or garden we can ask participants to simply take one or more pictures, and derive information from these.

Apart from improving measurement, using pictures might in some cases also decrease participation burden. For example, Wenz et al. (2020) asked respondents in the UK Household Longitudinal Study (UKHLS) to take pictures of purchase receipts, after which Optical Character Recognition (OCR) was used to produce a list of purchases as feedback to respondents. Potentially this not only eases the reporting task, but also offers positive feedback in return to respondents.

Despite the potential benefits, there are four broad challenges in implementing the collection of photos in survey methodology. (1) The first challenge is the willingness of participants to perform a task. Research so far shows that willingness differs by task and participant characteristics (Jäckle et al., 2019; Mulder & de Bruijne, 2019; Revilla et al., 2019; Struminskaya et al., 2021; Wenz et al., 2019). In the case of a housing survey, how willing are respondents to take a picture of their private space such as their own home? (2) The second challenge is ethics. Visual data can contain sensitive information, and ethical considerations are important both in the collection and analysis of pictures (Kelly et al., 2013). For instance, pictures could disclose personal information outside the main scope of the survey (Wenz et al., 2019). In the housing survey context, a picture might contain other household members or information that might disclose respondent's

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privacy such as a name or a bank card number. Thus, measures of privacy protection have to be designed and implemented. (3) The third issue is related to technical challenges in developing an app or web survey that allows the collection of pictures. The development and testing of an app or web survey can be time-consuming and expensive (Link et al., 2014). Moreover, other obstacles might occur such as an inability to develop apps applicable to all operation systems (Kreuter et al., 2018). Consequently, there is a risk that users with older or very new versions of an operation system cannot be included in the research. (4) The fourth challenge is related to the analysis of visual data. Since visual data are different than traditional survey data it requires a different analytical approach. The information of interest has to be detected, extracted and encoded from an image before the statistical analysis can be conducted (Schwemmer et al., 2022). One may rely on human coding for this task, but computer vision can improve the coding and most importantly accelerate this task (Bosch et al., 2019; Iglesias et al., 2022).

In sum, although there may be theoretical advantages of collecting photos in surveys more widely, there are several challenges that affect both the likelihood that data are collected and the quality of the resulting data. Apart from logistical issues, the central question is whether respondents are generally willing and able to take pictures and whether the potentially higher quality of the picture data can offset the potential problems of lower willingness and ethical issues.

To investigate the feasibility of using pictures in Web surveys, we describe an experiment conducted in the Longitudinal Internet studies for the Social Sciences (LISS) panel in the Netherlands. We invited panel members to participate in a Web survey about the characteristics of their dwelling. These participants, all in possession of a smartphone or tablet with camera, were randomly assigned to one of three experimental conditions: they were either asked to take pictures, answer text questions, or were given a choice to either take pictures or to answer survey questions.

To examine the overall feasibility of collecting visual data in a survey, we derived the following research questions: (1) How willing are participants to take pictures across these conditions? (2) What are the main reasons for not taking a picture and (3) Can we extract data of similar or even better quality from pictures than from survey questions? Answers to these questions form the stepping stone to further refinement of data collection and data analysis strategies.

The next section discusses earlier studies that aimed to collect picture data in surveys, and frames the problems with the willingness of people to participate and measurement in the larger Total Survey Error Framework, which we argue is a very useful framework for thinking about trade-offs in different types of error. After this, we introduce our experiment, which was designed to disentangle the different error components. We then discuss the compliance and quality of measurements across all the stages of data collection in our study and show that, although the different designs for collecting data lead to very different rates of compliance and quality of the measurements, the quality of our outcome statistic is similar across the conditions. We conclude by discussing ways forward for including the collection of pictures in surveys.

2 | BACKGROUND

2.1 | Visual data in survey research

Despite the fact that smartphones are widely used for answering surveys, there are few projects in which taking a picture using a smartphone was implemented as a supplementary task (Bosch

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et al., 2019; Jäckle et al., 2019; Link, 2013). According to Link et al. (2014) visual data can have four functions in survey research:

1. *Adding context to the survey research* – visual contextualisation of activities, locations or concepts asked in a survey. For example, taking a picture of the surroundings while answering a survey to provide context to the situation in which respondents complete a survey.
2. *Providing information that can be coded* – pictures can be used to replace complex (series) of survey questions with the goal of improving measurement quality.
3. *Improving respondent engagement with a survey* – Asking respondents to take a picture may increase compliance and engagement of participants in the survey.
4. *A training tool for respondents participating in long-term or more complex data-collection efforts* – this is related to providing video or photo instructions to respondents about how to collect data.

In some studies, the goals of using pictures are twofold. Link (2013), for example, tested compliance of participants to take a picture in a mobile longitudinal design study. A short mobile self-administrated survey was sent to participants five times a day via a mobile app. The study took place during the 5 weeks of the football World Cup in South Africa in 2010. At the end of each survey, participants were asked to take a picture and add a caption describing the picture. 66% of all the responses included a picture with a caption.

In the UKHLS (Wenz et al., 2020) visual data were used to investigate feasibility of scanning receipts to measure participant's spending. Panel members were asked to download an app on their smartphone or tablet. For a month, participants were instructed to report all their spending. This included sending scanned receipts, reporting spending without a receipt and non-spending days. Of all participants in the panel, 76% reported to own a mobile device. From them, 6% downloaded the app and used it at least once in the 5-week period (Jäckle et al., 2019). Information of interest were receipt-level, such as store, data and time of purchase, total amount, payment method, loyalty card use, and item-level such as product description, price and units bought (Wenz et al., 2020).

Bosch et al. (2019) tested whether it was feasible to use images in addition to a web survey. The target population in their study were people born between 1982 and 2003 from Mexico and Spain. Participants were asked to share two pictures along with answering the questionnaire regarding personality, usage of social media platforms, internet usage and demographic questions. The first photo was related to what they saw at that moment and second was a funny image already saved in the memory of their phone. 41% of respondents uploaded the image in line with the first task, and 49% uploaded the image in line with the second task. Bosch et al. (2019) then judged the content of the pictures using both manual and automated coding to judge the quality of the pictures, and showed that automated coding (object classification) was feasible.

In another study, Bosch et al. (forthcoming) used an opt-in panel to experiment with three different answer formats to survey questions: a text response, uploading an image, or uploading an image after receiving a motivational message. Their results show that the image conditions resulted in much lower response rates, most likely due to the increased burden, or situational aspects when the image questions are asked.

The studies just discussed all looked at compliance to take pictures. Struminskaya et al. (2021) showed that there often is a large difference between people's actual rates at which respondents submitted pictures (compliance) and stated willingness to perform extra tasks in smartphone surveys.

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Still, several studies that investigated hypothetical or stated willingness are interesting because they have focused more on understanding in what conditions respondents may be willing to perform tasks. Struminskaya et al. (2021) found that the general willingness to perform extra tasks in surveys using smartphones depends on the framing of the question, privacy concerns and familiarly using smartphones. Wenz et al. (2019) showed that participant characteristics also affect the willingness to take a picture. Persons with higher privacy concerns and less intense device usage reported lower willingness, with 65% of UKHLS panel respondents reporting to be willing to take a picture.

2.2 | Total survey error for survey questions verse pictures

For the comparison between pictures and question methods, we use the total survey error paradigm (TSE) (Groves & Lyberg, 2010), which has been used extensively to analyse different components of survey error, and has more recently also been used in situations where Big data are collected (Amaya et al., 2020; Jäckle et al., 2019). Central to the framework of TSE is the idea that a statistic is affected by two types of error: errors of measurement and representation.

One can make a further distinction into components of survey error that together make up representation error, and those leading to measurement error. Representation error combines errors in the selection and participation of respondents. The first component here is coverage error: how does the target population of the study align with the sampling frame that is used to select respondents for the study? The second component consists of sampling error: which elements are selected from the sampling frame to be selected into the sample, and finally non-response error: which elements who are invited for the study do actually participate?

Measurement error consists of three separate error components. First, does the concept that is measured align with the research objective (specification error), is the measurement itself accurate (measurement error) and finally, are measurements accurately processed into data that can be used to compute the statistic of interest (processing error). For a discussion of the TSE framework for survey questions we refer to Biemer (2010) for an overview. We here want to highlight how TSE components will differ when either survey questions or pictures are used for producing data.

We start with the measurement process. Formulating a task description for taking a picture is different than formulating a survey question. For both pictures and survey questions, researchers have to provide some guidance. For example, when we ask a survey question about the type and model of heating system respondents have in their home, we include text boxes in our survey so that respondents know what kind of information is needed. Further, researchers may include suggestions in a survey question for where to find this information, or examples of the kind of information that is desired. When respondents are asked to take a picture, researchers also need to include instructions on how to take the picture or what should and should not be in the picture. As we use the picture to derive the information we need, we need to tell respondents what kind of information (e.g. brand, type, serial number) is needed.

Error is also introduced in the process of doing the measurement itself. For example, in a survey question the respondent can misunderstand the instructions, or misremember the brand or type of the heating system. With a picture, the respondent may similarly misunderstand what is being asked or take a picture that is blurry, without sufficient light or contrast. But pictures will also contain much richer information about the variable of interest.

A key difference between collecting pictures and answers to questions is in the processing of information. Answers to survey questions are typically quite close to the statistic that we want

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to compute; little processing is needed to arrive at a statistic. This is not the case with a picture: even when pictures of a heating system are correctly taken, there remain several processing steps to arrive at a statistic. The desired information needs to be extracted from the picture, either manually or with the use of semi-automated computer coding.

On the representation side, the issue of selecting respondents using a sampling frame, and then drawing a sample, does not depend on the type of task we ask respondents to do. The key difference here is in the phase of nonresponse. Some respondents may be more or less likely to participate in a specific task, leading to differences in (item) response rates. There is a risk that asking respondents to upload a picture is seen as intrusive and burdensome, potentially leading to a lower willingness (Struminskaya et al., 2021) than asking them a survey question.

In sum, there is reason to believe that item-response rates are lower when we ask respondents to take a picture, rather than answer survey questions, leading to more representation error with pictures. However, in situations where it is complex to specify exactly what information we need, a picture may lead to more informative data yielding lower measurement error. Compared to survey questions, picture data would need to go through additional steps of data processing, in turn resulting in increased processing error, making it uncertain what the exact effect is on measurement error as a whole.

Previous studies have mainly investigated how pictures can be used to complement survey questions. In this study, we use an experimental design to investigate how textual survey questions or pictures lead to a different TSE structure. Within a survey, we assign respondents to provide data using either survey questions, or pictures, and compare the compliance and quality of data.

Because we embed the experiment in a survey, there are no differences in coverage, sampling error or unit non-response error between conditions. We expect item nonresponse error to our specific experimental questions due to willingness to be very small in the survey question condition, as respondents are here asked to perform a task that is common in the LISS panel. For the picture condition, item non-response error will be larger due to the fact that this is an unusual task for respondents. This may lead to an unwillingness to take pictures and/or technical issues.

For errors on the measurement-side of the TSE framework, we expect specification error to be similar across the conditions, as we will give respondents the same contextual information about the desired information. For measurement errors, we expect large differences: in the survey question, information may be incomplete because of memory problems or unclarity for respondents where to find the desired information. For the picture questions, we expect rich information, although pictures may suffer from problems such as being too dark or blurry. Overall, we expect measurement error to be lower in the picture condition. Finally, we expect processing error to be much larger in the picture condition, because the relevant information needs to be extracted from the pictures. For the survey questions, the answers by respondents should require little processing.

3 | METHODS

3.1 | Participants

To answer the research questions, we asked participants from the Dutch LISS panel to participate in a survey about their dwelling's characteristics. One month prior to this study, all panel members were asked whether they used a smartphone or tablet regularly and whether they were interested in participating in studies specifically on smartphones and/or tablets. We only selected respondents who said 'yes' to this open invitation to participate in smartphone-studies.

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The LISS[1] panel is managed by Centerdata, a research organisation based in Tilburg, the Netherlands. The panel has been recruited from samples drawn from the Dutch population register (Das & Knoef, 2019). Since its start, a number of refreshments samples have been included. The panel consists of approximately 7500 members from a total of 5000 households.

The study was conducted in October 2019. The collection of data started on the 7 October and finished on the 29 October, with a reminder sent on the 21 October. Only panel members with at least one smartphone or tablet were invited. From 2700 LISS panel members invited to participate, 1759 started the web survey (65%), and completed at least one task in which they were asked to answer questions or take pictures. One thousand seven hundred and thirty-five respondents completed the survey; the majority of dropout ($n = 24$) occurred at the final evaluation question of the survey. Individuals were randomly assigned to one of the following conditions:

1. ***Condition 1 (pictures)*** – within the web survey, respondents are asked to take pictures about three topics.
2. ***Condition 2 (survey questions)*** – participants answer survey questions.
3. ***Condition 3 (choice)*** – participants had the opportunity to choose either the picture or survey questions condition at the beginning of the survey. If participants agreed to take pictures, the routing was the same as for condition 1, and if they chose to answer survey questions, the routing followed condition 2.

3.2 | Survey questionnaire

The questions in the survey were inspired by the Netherlands Housing Survey (WoON). WoON is the survey about housing conditions of Dutch households. The questionnaire we used was programmed in Blaise 5.3 and QuestionnaireCTRL, and included the possibility to take pictures within the browser. We used the User Agent String of the device to detect whether respondents were using a mobile device (tablet or smartphone) to complete the survey. If respondents used a PC, they were shown a message prompting them to switch to a mobile device. They could not continue the survey on a PC.

The questionnaire contained five groups of questions, with the survey lasting about 10 min on average. See Appendix B for the flow of the questionnaire (Figure B1), and Appendix C for the full questionnaire that was used.

We asked participants in condition 1 and the condition 3 (those choosing to take pictures) to take three pictures. Before each task, we gave respondents a task description with a question asking if they wanted to complete it. If respondents said yes, they would first see a pop-up message asking users to give permission to the browser to operate the camera. After this, the phone or tablet would automatically switch to the camera-mode, allowing respondents to take a photo. If they rejected any of the three pictures at any point in the process, we asked an open-ended follow-up question about the reasons for being unwilling, followed by the survey question about the topic (these answers were then not used in the analysis) that was also asked to respondents in conditions 2 and 3.

In condition 2 and condition 3 (those choosing to answer questions), we asked participants to answer survey questions that measured the same concept of interest as the pictures.

[1]More information about the LISS panel can be found at: www.lissdata.nl.

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The three sets of pictures or survey questions included a picture/description of their favourite place in their house, the dwelling's outdoor area and heating system.

Favourite place in the home: We asked participants to describe by words or to take a picture of their favourite place.
Characteristics of home: This part was related to the description of the outside area of the home of respondents. This may have included a balcony, garden, terrace, courtyard, or patio. Only respondents whose dwelling had an outdoor space were additionally asked about its characteristics. In the survey route (condition 2 and those opting for survey questions in condition 3), we gave respondents two questions about the size of the outdoor area and how much of the outdoor area is paved. In the picture route, respondents were asked to take a picture of an outdoor area.
Heating system: In this part, participants were asked about the home heating system. Only participants who answered they had a central heating system were eligible for these questions. In the survey question condition (condition 2 and those opting for survey questions in condition 3) they were asked a text question about the brand and model of the heating system. In the picture route (condition 1 and those opting for pictures in condition 3), they were asked to take a picture showing the system including the brand and type. Again, if they rejected to perform a task, we gave them the survey question.
Evaluation: At the end of the questionnaire, we asked respondents who took at least one photo if they are willing to submit the pictures. Also, a short LISS survey evaluation instrument was given to everyone at the end of the survey.

3.3 | Analytic model

We will first focus on differences in representation error and answer the question whether people are willing to take pictures and answer survey questions, what the reasons are for (not) taking a picture across the three conditions (pictures, survey questions and choice). In order to do so we count the number of persons who were willing to take and send a photo or give an answer to survey questions. For the picture-conditions this corresponded with the number of pictures we obtained. For the question route, we counted the number of respondents who provided a response to the questions. This was done for participants in condition 2 and the condition 3 survey group. For open-ended questions, the respondents who gave answers including at least two or more characters were considered as willing to respond. For the outdoor space questions, only participants who answered both questions were considered as being actually willing. This also includes the response 'I don't know' for the question about garden size. To examine the reasoning behind a respondent's unwillingness to take a picture we code the responses to open-ended questions. One coder went through the answers and coded all the responses. If there was any uncertainty regarding the codes, a second person would also rate responses, and after discussion between coder 1 and 2, a category was chosen. We were generally interested in distinguishing respondents who were unwilling from those who were unable to take pictures. In the literature, a difference is often made between respondents willing to take a task, and those actually doing the task (or compliant). When we refer to willingness in this manuscript, we generally refer to respondents saying they want to do the task. In most cases, we are interested in those respondents actually sending us a picture, which we call compliance.

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We then turn to understanding the measurement process across conditions. We here chose to focus our analysis on one topic—heating systems. While the pictures of a favourite place and outdoor space contain much useful information, the nature of the information here is very different than the information contained in the text responses (pictures providing more context), making it hard to directly compare the extent of measurement quality of both sources. For the heating system, the text responses and pictures should capture the same information, and can therefore more easily be used to quantify the measurement quality.

There is a large variety of system in use in Dutch households that vary in how energy efficient they are. The goal of the current analysis was to establish the brand and model of heating system in use, so that at a later point in the housing survey this information can be used to derive indicators on the energy efficiency of the dwelling. Processing of the survey responses was done manually. For the survey questions the raw answers to the open-ended question about the heating system were used to classify the brand, model, build year and if available additional information about the heating system. For example, if text responses included only information of the build-year of heating system, then it would be coded as 'only a year'. For the pictures, a more elaborate processing scheme was used:

1. one researcher went through the pictures and masked the content that could lead to a violation of the privacy of the participants. In Appendix A we give an example of a picture and how the privacy control was done (Figure A1). In a second step, we manually went through the pictures and classified them in the following way:
2. we coded an image in line with a task if the respondent followed the instructions from the task description, and if both the brand and model of the system were visible. In most cases, the brand and model of the heating system are printed on the machine, but in some cases respondents took pictures of the manual, or just showed the complete system. These pictures were also deemed to be 'in line with the task'.
3. we pre-processed the images. Some pictures were somewhat blurry, noisy or skewed. This affects the extraction process (Gad, 2018). We adjusted the pictures' brightness and contrast where necessary so that they became easier to classify.
4. we then manually extracted the information of interest from the picture. Although we experimented with using OCR as a way to extract the information, in this paper we only report the results of manual coding the pictures. Appendix D (Tables D1 and D2) gives additional information on the information extracted by OCR and differences with manual coding of the pictures. One coder categorised brand, model, sub-type and build year for the systems. In cases where it was difficult to code these, a second independent coder was used. Codes were then discussed among the coders, and a decision was taken on whether the system could be classified (and to what degree).

4 | RESULTS

In this section we present the results of the analysis conducted to answer the research questions. First, we compare item/task non-response per picture. Further, we analyse the reasons for respondents' willingness to choose for pictures or survey questions, and actual compliance of sending a pictures or text answer. We compare the amount of information extracted from the pictures versus the survey questions for the topic of heating systems.

In total, 2700 LISS panel members were invited to participate in the survey. 65% ($n = 1759$) of invited panel members decided to fill in the survey, which results in a survey non-response

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TABLE 1 Participation in the study per experimental condition

	n	%
Invited	2700	100
Participated	1759	65.1
Survey non-response	941	34.9
Conditions		
Condition 1 (Pictures)	540	30.7
Condition 2 (survey)	655	37.2
Condition 3 (Choice)	564	32.1
Total	1759	100
Condition 3 choice		
Choice for pictures	320	56.7
Choice for survey	244	43.3
Total	564	100

TABLE 2 Compliance to take a picture or answer survey questions per condition

	Favourite place		Outdoor area		Heating system	
	n (eligible)	%	*n* (eligible)	%	*n* (eligible)	%
Condition 1 (Pictures)	212 (540)	39.3	138 (521)	26.5	155 (444)	34.9
Condition 2 (Survey question)	641 (655)	97.9	626 (632)	99.1	367 (549)	66.9
Condition 3 (Choice)						
Picture group	248 (320)	77.5	154 (306)	50.3	166 (276)	60.2
Survey group	233 (244)	95.5	228 (231)	98.7	117 (203)	57.6
Compliant picture 3[a]	248 (564)	44	154 (537)	28.7	166 (479)	34.7

[a]Compliant picture 3 – includes all the participants appointed to condition 3 regardless the choice they made between answering the survey question or picture.

of 35% ($n = 941$). At the start of the survey, respondents were randomly assigned to one of three conditions (condition 1: pictures, condition 2: survey questions, condition 3: choice). The somewhat higher number of respondents in condition 2 is due to chance. In condition 3, 57% ($n = 320$) of the participants chose to take pictures instead of answering survey questions. The complete distribution of the participants per condition is presented in Table 1.

4.1 | How willing are respondents to take a picture?

In Table 2, we present results on the compliance of taking a picture and answering text questions regarding a favourite place, a dwelling's outdoor space, and the dwelling's heating system.

Compliance in Table 2 corresponds to those respondents who successfully completed the task (be willing to take a picture, and actually take it). We find that compliance varies across

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the topics. Overall, participants show the highest compliance to take a picture of their favourite place in their home and the lowest compliance rate to take a picture of the dwelling's outdoor area. It is important to remember that the order of the questions was fixed, and for both survey questions and pictures we see a decreasing willingness the further respondents progress in the survey.

When we break down our results per condition and topic, we find that in condition 1, 39% of the participants took and submitted a picture of their favourite place. In condition 3 respondents were first asked to choose to either take pictures or answer survey questions. Of respondents who opted to take pictures in condition 3, 78% submitted a picture. When we also include respondents who opted to answer questions, 44% of all respondents in condition 3 submitted a picture.

For outdoor space we find again that submission rates are similar. Namely, 27% of respondents in condition 1 and 29% of all respondents in condition 3 (50% of group conditional on opting for pictures) decided to take and submit the picture of an outdoor space. Similarly, for a picture of a heating system, 26% of respondents in both condition 1 and condition 3 completed this task.

Regarding the compliance to survey questions around 99% of respondents in condition 2 and 99% in the survey group of condition 3 gave an answer to both questions about a dwelling's outdoor area. For the heating system, 67% (condition 2) or 58% (condition 3) were compliant .

When comparing compliance rates between condition 1 and the condition 2 we see a higher compliance to answer survey questions versus taking pictures across all the three topics. The highest difference was observed for the outdoor area: 27% of respondents in condition 1 completed the task of taking a picture while 99% of respondents in condition 2 gave an answer to text questions. The smallest differences were observed for the heating systems—35% of respondents in condition 1 took a photo while around 67% of respondents in condition 2 gave a response.

The differences in compliance between the picture and survey groups within condition 3 are smaller. For the heating system, respondents in the picture-group (condition 1) are similarly likely (60%) to take a picture than respondents in the survey group (58%) are to give a response.

4.2 | What are the main reasons for not taking a picture?

Respondents who did not submit a picture or text response, most often do this because they state they do not want to the task, rather than respondents indicating they would do the task, and then not submitting a picture (see Table A1 in Appendix A). For the pictures of the favourite place and heating system we asked respondents who did not agree to take a picture to provide the reason for not being willing to submit a picture using an open-ended question.

The coded responses to these open-ended questions are presented in Table 3. Lack of skills or malfunctioning of the device were categorised as a technical problem at the side of the respondent (e.g. broken camera, do not know how to upload a picture). Technical issues with the survey itself (e.g. unable to upload images, could not proceed with a survey after picture was taken) were coded as technical problems with the web survey. A certain number of respondents did not follow the instruction carefully and reported to be 'not at home' during the process of answering the survey. Some respondents also reported not to have a favourite place in their home. Finally, some respondents indicated 'other reasons' (e.g. messy house, inability to walk the stairs to take a picture, favourite place is occupied).

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TABLE 3 Coded reasons for unwillingness to take a picture of the favourite place and heating system in the home

	Favourite place		Heating system	
	n	(%)	*n*	(%)
Reasoning				
Unwilling				
Privacy concerns	202	50.5	58	14.5
Do not want or feel like it/unnecessary	26	6.5	14	3.5
Unable or incapable				
Technical problems with survey	40	10.0	39	9.8
Respondent's related technical issues	13	3.3	8	2.0
Do not have favourite place	20	5.0		
Unreachable[a]			196	49,1
Not sure how home is heated[a]			1	0.3
Not at home	13	3.3	11	2.8
Other reasons	17	4.3	15	3.8
Did not provide a reason	68	17.0	57	14.3
Total	400	100	399	100

Note: For dwelling's outdoor area picture, we did not ask respondents reasons for a rejection.
[a]Closed-ended answer-options given to respondents.

For a picture of the favourite place, 50.5% of respondents mentioned privacy reasons to explain their unwillingness to take a picture. Around 10% of respondents reported technical problems with the web survey. The most frequently reported reason regarding for not taking a picture of the heating systems was that the system is unreachable—49.1% of respondents reported this issue. Although it is possible that some heating systems are located in difficult-to-reach places, answering that a heating system is 'unreachable' may also serve as a 'soft' refusal. Consequently, it is hard to distinguish whether it is unwillingness or inability to perform a task. 14.5% of respondents reported privacy concerns while 9.8% of participants mentioned technical issues. After the questions asking why respondents refused to take a picture, they received as a follow-up the open-ended survey questions (not used in analyses). Here we find, that out of 1000 'no' answers to take a picture in condition 1, we received 879 answers to the follow-up survey question (87.9%).

4.3 | Can we extract more information of interest from pictures than from answers to survey questions?

One of the long-term goals of asking respondents about their heating system is to collect information about how energy-efficient dwellings are. To successfully do that, we need to at least identify the brand and model of heating system. We now turn to the question of whether we are able to derive this information from either the survey questions or pictures that were submitted.

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TABLE 4 Number of pictures and text responses with extracted information from the 321 submitted pictures and 484 submitted text responses of a heating system

	Pictures		Text	
In line with task	**Yes**	**No**	**Yes**	**No**
Number of pictures/ text responses	281	40	385	99
Privacy control	12	0	0	0
Pictures contain:				
Small part of a system without model information	0	33	—	—
Half or more of system including/excluding model specification	212	0	—	—
Only part of a system with system model specification	59	0	—	—
Includes information about system model (stickers)	8	0	—	—
Not informative/ not system of interest	0	7	—	—
Others (privacy censored picture)	2	0	—	—
Information extracted from the pictures[a]				
No brand or model information	38	12	0	78
Only brand of a system	101	28	243	0
Model of a system	124	0	92	0
More detailed model specification	18	0	50	0
Energy efficiency label or efficiency level	7	0	0	21

[a]Numbers do not add up to the sample size as categories for information from the pictures can overlap.

In total we obtained 321 pictures of heating systems. Table 4 shows that 4% ($n = 12$) of pictures were partly censored for privacy control. This means that pictures contained certain information that could disclose the identity of a participant. Only in one picture did this possibly censor the heating system label.

As a second step we judged whether pictures were in-line with the task description ('yes' or 'no' columns in Table 4). We found that 87.5% ($n = 281$) of the obtained pictures were in line with the task and showed (a part of) the heating system. Since we specified the task instructions broadly, respondents took different strategies in taking a picture, resulting in pictures containing different aspects of the heating system. In Table 4, we summarise what was shown on the picture. 66% of the pictures ($n = 212$) contained approximately half or more of the system. Pictures that were not in line with a task often contained a small part of the system (10.3%, $n = 33$). Further, around 2% of the pictures either contained heating systems that are not of our interest (e.g. radiators) or the pictures were not informative due to the lack of sufficient image quality to recognise the object (too dark, blurry).

In 44% ($n = 124 + 18$) of the 321 pictures submitted, we could manually detect information about the system model and type. Therefore, these pictures were of sufficient quality which made text labels visible and extractable. An additional 31% of pictures ($n = 101$) contained only the brand of the system, which to some degree is still useful information.

Text responses to the survey question about the heating system are summarised in the last two columns of Table 4. The ideal answer should here contain at least information on the heating system's brand and model. Around 29% ($n = 92 + 50$) of 484 text responses included this

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TABLE 5 Extent of non-response and measurement quality in the experiment extracting information about the brand and model of the heating system

	Pictures	Text	Test
Eligible to submit	720	752	
Response: Compliance (submitted)	321 (44.5%)	484 (64.4%)	$\chi^2(1) = 58.1, p < 0.01$
Measurement: quality ok (brand only)	240 (33.3%)	385 (51.2%)	Conditional on eligibility: $\chi^2(1) = 48.1, p < 0.01$ Conditional on compliance: $\chi^2(1) = 2.5, p = 0.11$
Measurement: quality good (brand + model)	139 (19.3%)	142 (18.9%)	Conditional on eligibility: $\chi^2(1) = 0.04, p = 0.83$ Conditional on compliance: $\chi^2(1) = 2.9, p = 0.09$

Notes: For details about the composition nonresponse error and measurement quality, see Tables 1–4. Samples (% of eligible respondents).

information, leading to a lower proportion of usable responses as compared to the pictures submitted.

The majority of the respondents were unable to provide a complete response that included all requested information. As a result, respondents used different strategies to answer the question partially. Most of these responses, 50% ($n = 243$), included only the brand of a heating system. 10% ($n = 78$) of responses included only information about the year a system was built, the energy source (e.g. 'gas') or other uninformative responses. Further, 4% ($n = 21$) of answers had only information on the type of a system (e.g. 'high-efficiency' or 'combination system'), but not any explicit brand or model of system. This information is more valuable than only information on the brand or year, but is still unusable in practice.

Table 5 summarises our results. While we find lower rates of non-response for respondents who answered survey questions as compared to the pictures, a series of χ^2 tests show that conditional on submitting a text or picture response, we find no overall difference in the quality of the submitted answers.

5 | DISCUSSION

In this study, we investigated the feasibility of respondents taking photos within a web survey. Theoretically, picture data can improve measurement accuracy while reducing the burden of participation. However, there are challenges in implementing the collection of visual data in addition to or instead of survey questions. These include the potential problem of unwillingness, ethical issues, technical challenges and the processing of the picture data into a statistic of interest. In our study, we directly compared pictures and text responses to understand the extent of representation and measurement error in each. Our analysis included examining actual willingness rates to take a picture versus answering a question, the reasoning behind the rejection of the task, and comparing the extracted information from pictures versus text responses in a survey about dwelling conditions.

We found that the compliance to take a picture varied between the three pictures we asked respondents to take. The highest compliance is for the task of choosing a favourite place in the home. 39% of respondents in condition 1 (picture) and 44% of respondents in condition 3 (choice) successfully completed this task. These compliance rates are rather high

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given that we anticipated this to be the most privacy-intrusive question. A reason for this may be that participants have control over the picture they want to share. For a picture of the outdoor area or heating system participants have less control over what they show and do not show. The lowest compliance was taking a picture of the dwelling's outdoor area, but even here, we find that giving people a choice between answering questions or taking a picture seems a good way to increase willingness rates. It is worth pointing out that the order of questions (favourite place, outdoor area, heating system) is important in interpreting our findings. One possible explanation why compliance rates are lower for the outdoor area and heating system is that respondents already went through the process of providing one or two pictures.

We generally observed that respondents were more willing to provide an answer to traditional survey questions than taking a photo over all three tasks. This means that using pictures as a measuring method might inflate the problem of non-response error.

Around 54% of respondents in Condition 3, who were given a choice between taking pictures or typing an answer to survey questions about the three central topics in the study, showed a preference to take pictures. This is similar to the willingness rates reported by Revilla et al. (2019) where 49% of participants indicated stated willingness to take and submit pictures by phone. Our findings indicate a slightly lower stated willingness when compared to the study by Wenz et al. (2019) where 65% of respondents said they were willing to use a camera to take a picture or scan a barcode. Further, we observed that in our study, stated willingness and compliance were strongly related. Only some respondents did not submit pictures after having indicated they were willing, and they mostly reported technical problems.

The main reason respondents said they were not willing to take a picture of the favourite place was privacy. Since we asked people to take a photo of their home, the concept we measured was potentially privacy-intrusive, and many respondents indicated that this was the main reason for not being willing to take pictures. This finding also corresponds with the literature where privacy is mentioned as a top reason for non-willingness (Jäckle et al., 2019; Revilla et al., 2019). Non-response due to privacy issues is not easy to prevent. However, it might help to provide respondents with more information about how the pictures will be stored, protected, and who will have access to data. For heating systems, 50% of respondents reported that their system was unreachable. Some respondents may have used this argument as a proxy for a soft refusal.

For content analysis we analysed one task—heating systems. We find that a large majority (87%) of the pictures were in line with the task. This means that most of the respondents understood and complied with the task description. Since the instruction given to respondents was quite general, what was in the pictures varied. From 44% of images we were able to extract the information of the heating system model. This means that those pictures had sufficient quality to manually find and code the information of interest. Further, we observed that these pictures also contained additional content which was not of our main interest (e.g. regular maintenance stickers). When respondents received the survey question about their heating system, respondents had to provide exact information about the brand and model type. This makes this question more cognitively demanding than other survey questions. Most of the respondents were unable to recall or find complete details. Around 42% of survey responses contain information on heating system brands while only 29% of respondents were able to recall or find details on the model and type, our information of interest.

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When we combine both representation and measurement quality, we conclude the following. In the picture conditions (including condition 1 and condition 3 picture group), 19% of all eligible respondents submitted pictures that allowed us to derive the information of interest. In the text questions (condition 2 and condition 3 text group), we find smaller representation error, but more measurement error, resulting in information of interest for 19% of all eligible respondents. This implies that both methods in this study lead to a similar data quality for the statistic of interest, although the error structures across the picture and text response conditions differ.

There are several limitations to the study. First, since we asked respondents to take a picture of a private area, we had to do additional protection of respondents. Consequently, we did not have background variables on respondents' characteristics which enable us to explain differences in respondents' compliance. Further, the study was conducted on the members of the LISS panel which means that respondents were more experienced in answering surveys and potentially more willing than the general population to take pictures. So, asking respondents to take pictures might be more challenging in official statistics where a sample of the general population is used.

It would be useful to additionally investigate willingness to take pictures in different settings that are not as privacy-intrusive as pictures of a home. The different task description can have an effect on how a participant takes a photo. More research should be done in understanding how different description relates to the way how participants take pictures. For example, if we show them an ideal photo, will they be more successful in taking it by themselves?

Another area of research is the reduction of processing and assessment of measurement error. Apart from manually coding the pictures, we have also explored text recognition (OCR) as a way to process information on the pictures automatically (see Appendix D). Although OCR would have helped us to reduce the amount of time in processing, we found that manual coding of the pictures led to more information to be extracted. However, improvements in the field of OCR may in the future perhaps allow us to even code pictures that were now to dark, or too blurry to code both manually, or using OCR. This would also allow us to quantify the extent of measurement quality more formally. The large majority of pictures and text responses were easy to categorise into one of the categories we used to extract information in terms of the brand and model of the system. We did however experience some responses that were more difficult to code, and quantifying measurement error using expert coding would have for allowed us to assess measurement error in a more detailed way. In terms of time, coding text questions was faster than coding the pictures, and the additional technical testing of the survey and questions from respondents result in higher costs when pictures are asked instead of questions.

Similarly, the pictures we have about respondents favourite place can be classified using object recognition, and the 'greenness' of the gardens can be coded using colour recognition. Using image recognition techniques can potentially lead to information being extracted quickly and at a low cost, while allowing respondents to report on concepts that are difficult to answer, or to provide context to survey questions.

In this study, we did not report on the accuracy of measurement for both the 'favourite place' and 'outdoor area' questions. The reason for this is that the survey questions and pictures for those two topics yield quite different information. An example description of the favourite place that we often encountered was 'my living room', while pictures would often depict a couch, chair or corner of the living room. Comparing the quality of measurements is here hard because the pictures

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give different information than the typed answers to survey questions, even when the instructions to respondents are the same. Here, pictures provide context and richness that answers to survey questions rarely—if ever—provide. A possible fruitful area that we recommend for future research is to use of pictures alongside the typed answers to survey questions to give context to survey responses.

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Data collection for this study was funded by Statistics Netherlands and Centerdata.

DATA AVAILABILITY STATEMENT

The survey data that support the findings of this study are openly available in the data archive of https://www.lissdata.org, reference number to be added later. The picture data are not publicly available. Data can be requested by contacting the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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APPENDIX A

TABLE A1 Type of task non-response per condition

	Favourite place		Outdoor space		Heating system	
	n	%	*n*	%	*n*	%
Condition 1 (Pictures)						
Reject the task	307	93.6	372	97.1	283	97.9
Did not submit	21	6.4	11	2.9	6	2.1
Total	328	100	383	100	289	100
Condition 3 (Choice)						
Photo group						
Reject the task	51	70.8	139	91.4	101	91.8
Did not submit	21	29.2	13	8.6	9	8.2
Total	72	100	152	100	110	100

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FIGURE A1 Example of privacy control. The blacked-out area contains address details. [Colour figure can be viewed at wileyonlinelibrary.com]

APPENDIX B. STUDY FLOW

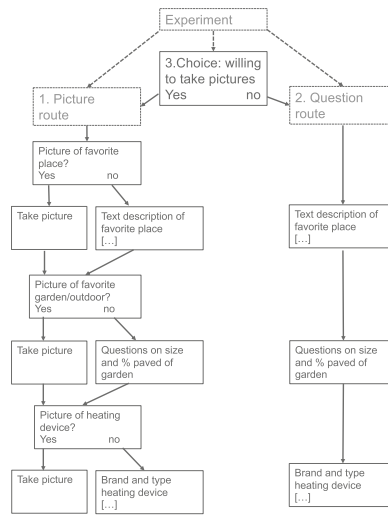

FIGURE B1 Overview of the study showing the routing of the questionnaire (see Appendix C), and the questions where respondents were either asked to take a picture or answer a question, depending on the experimental condition they were assigned to, and answers respondents gave in the 'choice' condition

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APPENDIX C. QUESTIONNAIRE

C.1 Introduction

This is a questionnaire about your home, carried out by researchers from Statistics Netherlands and Utrecht University. In this questionnaire, we test a number of new ways to measure characteristics of your home. We would for example like to know how you heat your home, how your home looks like, and what your favourite part of your home is.

There are two important things for this study:

1. Please complete this questionnaire at home. If you are somewhere else at the moment, please login in at a later moment
2. [EXPERIMENT with task description (conditions 1–3, see below)]

[Condition1]: Please use your smartphone or tablet, so that we can ask you to take pictures about certain topics
[Condition2]: Please use your smartphone or tablet
[Condition3]: Please use your smartphone or tablet, so that we can ask you to take pictures about certain topics. You can choose per topic whether you would prefer to take a picture or complete questions instead.

Please click 'next' to continue
[Condition1 = exp1 from here on; Condition1 = exp2; Condition3 = exp3].

SECTION: GENERAL
[CHECK: does the user use a smartphone or tablet [If not Q0].

Q0. It appears that you are using a laptop or PC to complete this questionnaire. Is that correct?

1. Yes
2. No -> Q1a.

[If Q0 = yes].
You can only complete this questionnaire on a smartphone or laptop. If you do not have a smartphone or laptop, you can now quit this questionnaire.

[All respondents].
Q1a. How satisfied or unsatisfied are you with your home?
1. Very unsatisfied.
2. Unsatisfied.
3. Neither satisfied, nor unsatisfied.
4. Satisfied.
5. Very satisfied.

Q1b. To what extent do you feel attached to your home?
1. Not attached at all.
2. Not attached.
3. Attached.
4. Very attached.

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[if exp = 3 ASK Q2.]
Q2. In this study, we would like to ask you several times to take pictures of parts of your home. You can also choose to answer questions, but your answers are then less valuable for this study. You can make a choice with every question to either take a picture or not.
Are you willing to take several pictures?

1. Yes
2. No

[if exp = 1 OR Q2 = YES ASK Q3.]
Q3. Many people have a part of their home, or multiple parts where they really like to be. Would you be willing to take a picture with the camera on your smartphone or tablet which you use to complete this questionnaire of one of your favourite places in your home?

1. Yes [-> take pic -> Q.4]
2. No

[IF Q3 = 2.]
Q3a. You did not want or was not able to take a picture of your favourite part of your home for this study. What is/are the most important reason(s) for this?
(Large text box)
[If exp = 2 or Q2 = No or Q3 = No ASK Q3b.]

Q3b. Many people have a part of their home, or multiple parts where they really like to be. Could you give us a description of one of your favourite parts of your home? How does this part look like?
(Large text box)

Q4. The next questions are about some of the characteristics of your home.
How many rooms does your home have?
Please include living room(s), bedrooms and study rooms that you use yourself.

(Small text box)
Q5. Do you have a garden, patio, courtyard or field that belongs to the house?
Multiple answers possible.
1. Garden.
2. Patio.
3. Courtyard.
4. Field.
5. Terrace/balcony.
6. None of these.
[if Q5 = 1–5 AND (exp = 1 or Q1 = yes) ASK Q6a].

Q6a. We are interested in measuring to what extent your garden consists of terraces, stones or plants or grass. Will you to use your camera take a picture of your outside space? (Garden, courtyard, patio, terrace, balcony)?
In case you have a garden at multiple sides of your home, please take a picture of that part of your garden where you spend most of your time.

1. Yes
2. No

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[if Q5 = 1–5 OR no photo taken ASK Q6a Q6b ELSE ASK Q7].
Q6b. What is the approximate size (in square metres) of your own garden, patio, courtyard or terrace put together?
If you are unsure, please give us an estimate.
(Small text box) with label m2.

3. Do not know

Q6c. What part of your outside space is paved? If you are unsure, please give us an estimate

1. Almost nothing
2. A small part
3. About half
4. Most of the outside space
5. Almost all of the outside space

SECTION: ENERGY

Q7. The next questions are about the ways in which you hear your home, and the system you use for that.
1. Own heating system (CV, or water pomp).
2. Block or neighbourhood heating-> q9.
3. City heating -> q9.
4. Heating by a fireplace or heater -> q9
5. In another way-> q9.
6. Do not know.
[if (exp = 1 or Q2 = Yes) ASK Q8a] else -> 8c.

Q8a. We would like to know how efficient your heating system is. Will you take a picture of your CV/heating system? Please take this picture in such a way that the brand and type is visible, or otherwise as much of the heating system as possible.

1. Yes -> photo -> Q9
2. No

Q8b. You did not want or was not able to take a picture of your CV/heating system. What are the most important reasons for this?

1. I cannot reach my heating system
2. I am not sure how my home is heated
3. Other, … [small textbox]

[if Q8a = 2 or (exp = 2) or (exp = 3 and Q2 = No)].
Q8c. Could you write down the brand, type and build year of your CV/heating system? In case you do not know the build year, you can leave this information out. You can also leave the entire answer box empty
(text box)

Q9. [If (Q3a = Yes or Q6a = Yes or Q8a = Yes) We are almost at the end of the questionnaire. You have taken one or multiple pictures of parts of your home. Would you like to share these pictures with researchers of SN and CentERdata?

1. Yes, I want to share these pictures
2. No, I do not want to share these pictures

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SECTION: END/evaluation
[LISS standard evaluation questions].
Q10a. Did you find it easy or difficult to answer the questions in this study?
1. Very easy … , 5. Very difficult [end-point labels only]

Q10b. How interesting did you find the topic of this questionnaire?
1. Very interesting– not interesting at all [[end-point labels only]
[if exp = 1 or 3].

Q10c. Did you experience any technical difficulties with this questionnaire?
1. Yes
2. No

[if exp = 1 or 3].

Q10d. Did you find that taking pictures violated your privacy?
1. Yes
2. No -> Q11
[if exp = 1 or 3].

Q10e. We asked you to take pictures of your favourite part of your home, your garden and your heating system. Can you explain which of these questions most privacy sensitive to you was?

Q11. Finally, do you have any other remarks about this questionnaire? We are interested in your opinion.
(Large text box)

APPENDIX D. OBJECT CHARACTER RECOGNITION OF PICTURES

TABLE D1 Results of optical character recognition (OCR) model to detect label information of a heating system

	n	%
Image included in the analysis	142	100
Images with detected text	140	98.6
Images with detected and recognised text	139	97.9
Extracted by OCR		
Model and brand of a device		
No difference in characters	44	31.0
<4-character differences	42	29.6
Model of a device		
No difference in characters	6	4.2
<4-character differences	2	1.4
Brand of a device		
No difference in characters	14	9.9
<4-character differences	3	2.1
Not meaningful text extracted	28	19.7
Not text detected	2	1.4
Fewer than two characters	1	0.7
Total	142	100

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TABLE D2 Match between optical character recognition (OCR) model and human (manual) coding

	n	%
Image included in the analysis	142	100
Comparing two methods		
Match between OCR and manual coding		
No difference in characters	44	31.0
<4 characters difference	37	26.1
Total	81	57.0
No match	61	43
Total	142	100

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